

EDM v2 - Go-Live SOP (Technical)

Σκοπός: Technical Go-Live Standard Operating Procedure για το EDM v2 — prechecks, εγκατάσταση, configuration, Hermes routing και appendices με δείγματα configuration και λειτουργικότητες της ολοκληρωμένης έκδοσης.

Σύνοψη

Controlled production rollout για το EDM v2 με βασικά components: Docker, PostgreSQL (+pgvector), MinIO, Redis, FastAPI backend, parsing tools (Tesseract, pdfminer/camelot/tabula), local LLM runtimes, Hermes agent, CI/CD, monitoring, backups.

Preconditions / Απαιτήσεις

- GitHub repo για EDM (έτοιμο).
- Plesk domain (π.χ. pylon.ergalyon.com) με SSH access και credentials.
- Server (Hetzner VM) με root / sudo access ή Plesk admin.
- Laptop developer με admin privileges.
- Λίστα credentials (GitHub token, MinIO access, DB password) σε secure vault.
- Pilot supplier: Poimenidis (sample data).

Laptop — developer machine

Εγκαταστήστε και ελέγξτε τα ακόλουθα:

Βασικός έλεγχος συστήματος

```
uname -a
df -h
free -h
```

Docker + docker-compose (Ubuntu example)

```
sudo apt update
sudo apt install -y ca-certificates curl gnupg lsb-release
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearm
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings
sudo apt update
sudo apt install -y docker-ce docker-ce-cli containerd.io
docker compose version
sudo usermod -aG docker $USER
```

Python & Virtualenv

```
sudo apt install -y python3 python3-venv python3-pip
python3 -m venv .venv
source .venv/bin/activate
pip install --upgrade pip
pip install fastapi uvicorn[standard] pydantic sqlalchemy alembic psycopg2
```

Tesseract & PDF tools

```
sudo apt install -y tesseract-ocr tesseract-ocr-eng default-jre
pip install pdfminer.six camelot-py[cv] tabula-py pillow
```

Local LLM runtime

Επιλέγεται ανάλογα hardware: Ollama (user-friendly), llama.cpp (CPU), TGI/vLLM (GPU).
Εγκαταστήστε/ρυθμίστε το κατά περίπτωση.

Docker-compose test

Τρέξτε ένα minimal docker-compose (Postgres, MinIO, Redis) και ελέγξτε τα containers:

```
docker compose up -d
docker ps
docker logs <container>
```

Server (Plesk / Hetzner)

Bootstrap production server με Docker-based deploy. Βασικά βήματα:

- apt update && apt upgrade -y
- Install Docker & docker-compose plugin (όπως laptop)
- Configure Plesk deployment or GitHub deploy hooks
- Deploy docker-compose stack: postgres (+pgvector), redis, minio, web (FastAPI), traefik/nginx, prometheus, grafana
- Enable TLS (Let's Encrypt), configure firewall (UFW)

PostgreSQL + pgvector

Χρησιμοποιήστε image που υποστηρίζει pgvector ή custom image. Μετά τη δημιουργία του container, ενεργοποιήστε το extension:

```
CREATE EXTENSION IF NOT EXISTS vector;
```

MinIO

```
docker run -d --name minio -p 9000:9000 -e MINIO_ROOT_USER=minioadmin -e M
```

Redis

```
docker run -d --name redis -p 6379:6379 redis:7
```

Reverse Proxy & TLS

Traefik ή NGINX. Configure ACME/Let's Encrypt via Plesk or Traefik. Ensure ports 80/443 open.

Backups

DB backup via cron + pg_dump (or pgBackRest). MinIO snapshots via mc to external object storage. Test restores.

Monitoring

Deploy Prometheus & Grafana containers. Scrape /metrics from services. Configure Alertmanager.

Application-level configuration

- Store secrets in Vault or CI/CD secrets; never commit secrets.
- Use Alembic/Django migrations; test migrations in staging.
- Provide /health and /metrics endpoints.

Hermes agent & Model routing setup

Hermes is a FastAPI-based routing service that evaluates policy rules (YAML) and routes tasks to local models, freeLLMAPI, Abacus, or Review Queue. It integrates with Redis queues and the model registry.

Policy file example

See Appendix A for a sample `routing_rules.yaml`.

Tests / Validation (pre-go live)

- End-to-end tests with Poimenidis sample invoices (parse → normalize → match/review → export).
- Security tests: TLS, firewall, RBAC, ensure unauthenticated API calls return 401/403.
- Backup & restore test: restore DB dump in test instance and validate.

Go-Live Rollout Plan (pilot)

1. Deploy to staging and run full validation with Poimenidis samples.
2. Schedule production deploy during low traffic window.
3. Enable only Poimenidis processing; keep review queue mandatory.
4. Monitor metrics & logs for 24–72 hours.
5. Gradually add more suppliers if stable.

Acceptance Criteria

- Core workflow passes end-to-end for Poimenidis (parse → normalize → match/review → export).
- Backups tested and recoverable.
- Monitoring & alerts established and tested.
- Review UI functional and human workflows validated.
- Security & TLS verified.
- Rollback process documented & tested.

Appendix A — Sample configuration files

A.1 docker-compose.yml (ενδεικτικό)

```
version: '3.8'

services:
  postgres:
    image: postgres:15
    environment:
      POSTGRES_PASSWORD: example
    volumes:
      - postgres_data:/var/lib/postgresql/data
    ports:
      - '5432:5432'

  redis:
    image: redis:7
    ports:
      - '6379:6379'

  minio:
    image: minio/minio:latest
    command: server /data
    environment:
      MINIO_ROOT_USER: minioadmin
      MINIO_ROOT_PASSWORD: minioadmin
    ports:
      - '9000:9000'
```

```

web:
  build: .
  command: uvicorn app.main:app --host 0.0.0.0 --port 8000 --workers 1
  ports:
    - '8000:8000'
  environment:
    - DATABASE_URL=postgresql://postgres:example@postgres:5432/ergalyon
    - MINIO_ENDPOINT=minio:9000
    - REDIS_URL=redis://redis:6379/0
  depends_on:
    - postgres
    - minio
    - redis

traefik:
  image: traefik:v2.9
  command:
    - '--api.insecure=true'
    - '--providers.docker=true'
    - '--entrypoints.web.address=:80'
    - '--entrypoints.websecure.address=:443'
  ports:
    - '80:80'
    - '443:443'
    - '8080:8080'
  volumes:
    - /var/run/docker.sock:/var/run/docker.sock:ro

volumes:
  postgres_data:

```

A.2 routing_rules.yaml (Hermes sample)

```

# Hermes routing rules (YAML)
# Each rule evaluated in order. Fields: name, condition, action, fallback
-
  name: 'sensitive_tax_data'
  condition: "task.contains('tax_id') == true"
  action: 'local'
  reason: 'Never send tax IDs externally'
-
  name: 'low_latency'
  condition: "task.sla_ms < 300"
  action: 'local'
  reason: 'Prefer local small model for low latency'

```

```

-
  name: 'complex_rag'
  condition: "task.type == 'rag_search' and local_model.vector_size >= 160
  action: 'local'
  fallback: 'abacus'
-
  name: 'default_fallback'
  condition: 'true'
  action: 'freeLLMAPI'
  fallback: 'abacus'

```

A.3 .env.example

```

# .env.example
DATABASE_URL=postgresql://postgres:example@postgres:5432/ergalyon
MINIO_ENDPOINT=http://minio:9000
MINIO_ACCESS_KEY=minioadmin
MINIO_SECRET_KEY=minioadmin
REDIS_URL=redis://redis:6379/0
SECRET_KEY=replace_with_strong_random
PGVECTOR_EXTENSION=true

```

Appendix B — Λειτουργικότητες ολοκληρωμένης έκδοσης

- Complete intake: PDF/XML/Excel upload, supplier detection, source_file_id tracking.
- Deterministic parsing core + supplier-specific normalization (e.g. Poimenidis 03-prefix).
- Multi-layer parsing: deterministic → rule-based → local AI extractors → freeLLMAPI/Abacus fallback.
- RAG-backed Supplier Memory & Document Store (MinIO + pgvector retrieval).
- Product model with ProductSupplierLink, provenance, price history, provisional flags.
- Review Queue UI with side-by-side diffs, accept/reject/merge, bulk actions, keyboard shortcuts.
- Hermes routing agent (policy-based) with model registry, health checks, logging.
- Model governance: nightly eval harness, labeled test set, drift detection, metrics.
- CI/CD + GitOps (GitHub Actions), IaC (Terraform for Hetzner), containerized deploys.
- Monitoring & alerting (Prometheus, Grafana, Alertmanager), structured logs (OpenSearch/ELK).
- Backups & recovery (pg_dump/WAL, MinIO snapshots), tested restore processes.
- Secrets management (Vault or equivalent), RBAC (Keycloak) and audit trails.
- Supplier Rule Engine UI for non-devs (regex rules, normalization tests).
- Enrichment pipeline: catalog import → manual attachments → scraping (Scrapy/Playwright) as last resort.

- Export module with validation & preview for Pylon ERP format (excel/xml).
- Feature flags & canary releases for risky automations.
- Automated tests: unit, integration, e2e with real sample corpus (Poimenidis).
- Operational runbooks, incident response, SLOs and error handling.
- Optional: on-prem GPU inference, vector DB (Weaviate/Milvus) if scale requires.

Παραδοτέα: docker-compose.yml (dev/staging), .env.example, routing_rules.yaml (Hermes), setup_scripts.md, GitHub Actions CI template, minimal seed data (Poimenidis samples).

Αν θέλεις, μπορώ να επισυνάψω και ένα ξεχωριστό αρχείο με τα scripts/αρχεία έτοιμα για το repo (docker-compose.yml, routing_rules.yaml, .env.example) ως ξεχωριστά αρχεία. Θες να τα προσθέσω στο repo ή να τα κατεβάσεις τώρα ως αρχεία;